RA-6200 6-Axis Industrial Robot Arm

Technical Reference & Maintenance Manual

System Designation: RA-6200 **Controller Version:** RC-7 Series **Manual Revision:** 4.2 **Release Date:** November 2023 **Applicable Units:** All RA-6200 units manufactured from Serial No. RA62-2300 onwards

---

> **SAFETY NOTICE — READ BEFORE PROCEEDING** The RA-6200 is an industrial robot capable of high-speed, high-force motion across a large work envelope. Serious injury or death can result from improper operation, maintenance, or safety system bypass. All personnel working with or near this system must complete the mandatory safety induction before operating or entering the robot work cell. This manual does not replace formal safety training.

System Overview

The RA-6200 is a six-axis articulated industrial robot designed for material handling, assembly, machine tending, and palletising applications in manufacturing environments. It combines a 20 kg payload capacity with a 1,850 mm reach radius and a compact base footprint, making it suitable for installation in constrained production cells.

The robot is controlled by the RC-7 Series controller, which supports teach pendant programming, offline programming via PC software, and remote I/O integration with PLCs and factory automation systems. The system is designed for 24/7 continuous operation with scheduled preventive maintenance intervals.

---

1. Axis Configuration & Joint Specifications

The RA-6200 uses a standard 6-axis articulated configuration. Axes are numbered J1 through J6 from base to wrist.

Axis	Description	Motion Range	Max Speed
J1	Base rotation	±170°	195°/sec
J2	Lower arm pitch	+85° / –95°	175°/sec

J3	Upper arm pitch	+70° / –185°	185°/sec
J4	Wrist roll	±190°	250°/sec
J5	Wrist pitch	±120°	250°/sec
J6	Wrist rotation (tool flange)	±360°	400°/sec

2. System Specifications

Mechanical:

Parameter	Value
Payload capacity	20 kg
Maximum reach	1,850 mm
Repeatability	±0.05 mm
Robot weight (arm only)	285 kg
Base mounting	Floor, ceiling, or angled mount
Ingress protection	IP54 (standard), IP67 (wash-down option)
Operating temperature	0°C – 45°C
Relative humidity	20% – 80% RH (non-condensing)

Controller (RC-7):

Parameter	Value
Controller dimensions (W×D×H)	600 × 550 × 1,000 mm
Power supply	3-phase, 380–480V / 50–60Hz
Power consumption (peak)	4.5 kW
Communication interfaces	EtherNet/IP, PROFINET, DeviceNet, RS-232
Program storage	10,000 positions, 9,999 program steps
Teach pendant display	8.4 inch colour touchscreen
Emergency stop circuits	Dual-channel, Category 3 / PLd

3. Normal Operating Conditions

The following parameters represent healthy system operation. Deviations should be investigated before continuing production.

Parameter	Normal Range
Joint temperature (all axes)	Below 65°C
Controller internal temperature	Below 55°C
Servo drive regenerative load	Below 70%
Brake release time (any axis)	Below 150 ms
Position deviation at rest	Below 0.02 mm
Payload during operation	At or below 20 kg
TCP speed (during production moves)	At or below programmed speed limit
Battery voltage (program backup)	3.0V – 3.6V

---

4. Fault Codes & Troubleshooting

Fault Code F-01 — Joint Overtemperature

Subsystem: Servo / Mechanical **Severity:** Major — robot halts immediately

Description: One or more joint servo motors or gear reducers has exceeded the safe operating temperature of 65°C. The controller shuts down servo power to all axes and holds the robot in its current position using brakes.

Common causes:

- Cycle time too aggressive — insufficient cooling time between moves at high speed and payload
- Ambient temperature in the work cell is elevated beyond specification
- Grease degradation in the gear reducer — increased friction generating excess heat
- Servo drive cooling fan failure inside the controller cabinet

Diagnostic steps and resolution:

1. Record which joint triggered the fault — the fault detail screen on the teach pendant shows the specific axis and temperature reading. Navigate to: Alarm → Fault Detail → Joint Temperatures.

2. Do not attempt to restart immediately. Allow a minimum 30-minute cool-down period with the servo power off but the controller powered on (fans continue running).

3. After cool-down, check the ambient temperature in the work cell using a thermometer. If ambient is above 40°C, the work cell ventilation is insufficient — contact facilities before resuming.

4. If ambient is within range, review the robot program cycle time. Calculate the duty cycle — the ratio of move time to total cycle time. If duty cycle exceeds 80% at maximum speed and payload, reduce speed by 15% or add a dwell time between moves.

5. Check the controller cabinet cooling fans — open the rear panel of the RC-7 controller (power off first, lock out the main isolator). Confirm all fans are spinning when power is restored. A failed fan must be replaced before resuming operation.

6. If the fault recurs within 2 hours of normal operation after the above checks, the gear reducer grease in the affected joint may have degraded. This requires a grease replacement service — contact the equipment engineering team. Do not continue production with recurring overtemperature faults.

Fault Code F-02 — Position Deviation Excessive

Subsystem: Servo Control **Severity:** Major — robot halts immediately

Description: The difference between the commanded position and the actual encoder-measured position for one or more axes has exceeded the allowable deviation threshold. This indicates the robot is not following its programmed path accurately.

Common causes:

- Mechanical collision — robot arm has contacted a fixture, workpiece, or cell structure
- Payload exceeding rated capacity — servo cannot maintain position under load
- Encoder cable fault — intermittent signal loss causing position tracking error
- Servo drive fault — drive is unable to deliver required torque

Diagnostic steps and resolution:

1. Inspect the work cell visually before doing anything else. Determine whether a collision occurred — look for damage to the robot arm, tooling, fixtures, or workpieces in the cell.

2. If a collision occurred: do not attempt to move the robot under power until the arm has been visually inspected for damage. Check all six joints for structural damage, cracking, or deformation. If any damage is found, tag the robot out of service and

contact the equipment engineering team.

3. If no collision occurred: check the payload. Weigh the gripper and workpiece together — the combined weight must not exceed 20 kg at the tool flange.

4. Navigate to: Diagnostics → Servo → Position Deviation Log. Review the deviation values for each axis at the time of the fault. A single axis with high deviation suggests a mechanical or encoder issue on that joint. Multiple axes with simultaneous deviation suggests a program or payload issue.

5. Check the encoder cable for the affected axis — follow the cable from the joint to the controller. Look for pinching, chafing, or loose connections at both ends.

6. If all checks are clear, clear the fault and run the program at 30% speed. Gradually increase to full speed over 3 cycles while monitoring the deviation log.

Fault Code F-03 — Brake Fault

Subsystem: Safety / Mechanical **Severity:** Critical — robot halts immediately, brakes engaged

Description: The brake release time for one or more axes has exceeded 150 ms, or the brake has failed to engage or release on command. The brake system is a critical safety component — this fault requires immediate attention and must not be cleared without investigation.

Common causes:

- Brake coil degradation — increased resistance reducing release force
- Brake disc wear — physical wear causing inconsistent engagement
- Brake power supply voltage out of specification

Diagnostic steps and resolution:

1. Do not clear this fault and resume operation without completing this procedure in full.

2. Navigate to: Diagnostics → Safety → Brake Test. Run the automated brake test sequence. The test applies and releases each brake individually and measures engagement time. Record the result for each axis.

3. Check the brake power supply voltage in the controller cabinet — must be 24V DC ± 10%. If voltage is out of range, the power supply module requires adjustment or replacement.

4. If the brake test reports a specific axis as failing: the brake assembly on that joint requires inspection by the equipment engineering team. The robot must not be

returned to automatic operation until the brake on the affected axis is confirmed functional.

5. As a temporary measure, the robot may be operated in manual reduced-speed mode (below 250 mm/s TCP speed) with a trained operator present and the work cell cleared of all other personnel — this is only permitted if the equipment engineering team has been notified and has provided written approval.

Fault Code F-04 — Controller Temperature High

Subsystem: Controller Electronics **Severity:** Major — robot performance reduced, shutdown imminent if unresolved

Description: The RC-7 controller internal temperature has exceeded 55°C. Continued operation above this threshold risks damage to servo drives, I/O modules, and the main processor board.

Common causes:

- Controller cabinet cooling fans failed or running below speed
- Cabinet air filter clogged — restricted airflow
- Ambient temperature in the controller location exceeds specification
- High servo regenerative load generating excess heat inside the cabinet

Diagnostic steps and resolution:

1. Check the controller cabinet air filter — located on the lower front panel behind a removable cover. If the filter is visibly grey or clogged, clean it immediately: remove the filter foam, rinse under clean water, allow to dry fully, and reinstall.

2. Check that all cabinet fans are running. With the cabinet powered on, open the rear panel and listen for fan noise. Confirm visually that all fans are spinning. Replace any failed fans with the correct replacement (part number RA6200-CF-006).

3. Ensure the controller cabinet has at least 300 mm of clear space on all sides for airflow. Do not place objects against the cabinet vents.

4. If the ambient temperature near the controller exceeds 40°C, the installation environment is outside specification. Contact facilities to improve ventilation.

5. If the cabinet temperature continues rising above 60°C after the above checks, shut down the system and contact the equipment engineering team before restarting.

Fault Code F-05 — Safety Zone Violation

Subsystem: Safety System **Severity:** Critical — robot halts immediately, requires manual reset

Description: The robot has entered or attempted to enter a defined safety zone in the work cell configuration. Safety zones are defined in the controller software and represent areas where robot motion is restricted to protect personnel, fixtures, or adjacent equipment.

Common causes:

- Program path deviation due to a preceding fault causing incorrect positioning
- Safety zone configuration was modified and the program has not been updated to match
- Operator entered the work cell during automatic operation triggering the area scanner

Diagnostic steps and resolution:

1. Confirm the work cell is clear of all personnel before any reset attempt. Physically inspect the cell — do not rely on indicators alone.

2. Navigate to: Safety → Zone Violations → Last Event. Review the zone that was violated, the axis positions at the time of violation, and whether the trigger was a programmed path or a physical sensor (area scanner, light curtain).

3. If triggered by an area scanner or light curtain: someone entered the work cell during operation. This is a serious safety event. Record the incident and review with the safety officer before resuming production.

4. If triggered by a program path deviation: the robot deviated from its expected path due to a preceding fault. Review the fault log for any faults that occurred before F-05. Resolve the root cause fault first.

5. If triggered by a configuration mismatch: check whether the safety zone parameters were recently modified. If so, the robot program must be re-verified against the new zone boundaries by a qualified programmer before resuming.

6. Manual reset requires a key-switch operation on the teach pendant followed by confirmation on the RC-7 controller panel. This two-step reset is intentional — it cannot be cleared remotely.

Fault Code F-06 — Program Checksum Error

Subsystem: Controller Software **Severity:** Major — program cannot be executed

Description: The robot program file stored in the controller has failed its integrity check. The program may have been corrupted during a save operation, a power interruption, or a controller firmware update.

Common causes:

- Power interruption during program save
- Controller firmware updated without backing up programs first
- USB transfer corruption

Diagnostic steps and resolution:

1. Navigate to: Programs → Program List. Identify the program showing the checksum error.

2. Check whether a backup copy of the program exists on the USB drive or the offline PC programming software. If a backup exists, reload the program from backup.

3. If no backup exists, the program must be recreated or re-taught. Contact the process engineer responsible for the program — do not attempt to run a partially corrupted program.

4. To prevent recurrence: after any program modification, always save to both the controller memory and a USB backup. This is standard procedure and must be followed after every programming session.

Fault Code F-07 — Backup Battery Low

Subsystem: Controller **Severity:** Minor — operation continues, maintenance action required

Description: The backup battery voltage in the RC-7 controller has dropped below 3.0V. This battery maintains all program data and position registers when the controller is powered off. If the battery fails completely, all programs and positions are lost on the next power cycle.

Common causes:

- Battery has reached end of service life — typical battery life is 3 to 5 years
- Controller has been powered off for an extended period — battery depletes faster without periodic charging from the controller

Resolution Steps:

1. This fault does not require immediate shutdown — production can continue. However, battery replacement must be scheduled within 2 weeks.

2. Before replacing the battery, ensure the controller is powered on — the battery replacement procedure must be performed with the controller live to prevent data loss. Refer to the battery replacement procedure in Appendix D.
3. Replace with the correct battery type only — part number RA6200-BB-007. Using an incorrect battery type can damage the controller board.
4. After replacement, verify that all programs and position registers are intact by navigating to: Programs → Program List and confirming the program count matches the pre-replacement record.

---

5. Preventive Maintenance Schedule

Daily (Before First Shift)

- Visually inspect the robot arm for any signs of impact damage, oil leaks at joints, or loose cable management
- Confirm all safety devices are functional — test the emergency stop, safety gate interlock, and area scanner before enabling automatic mode
- Check teach pendant cable and connector — no damage or strain
- Run the standard startup self-check sequence from the teach pendant and confirm all axes home correctly

Weekly

- Clean the controller cabinet air filter
- Inspect all external cables on the robot arm — cable management clips, drag chains, and conduit entries
- Check the tool flange and end effector mounting — all bolts must be at specified torque
- Run the brake test sequence (Diagnostics → Safety → Brake Test) and record results
- Lubricate the tool flange bearing with the specified grease if cycle count exceeds 50,000 since last lubrication

Monthly

- Check joint temperatures during a full production cycle using the temperature monitoring display
- Inspect all axis cable harnesses inside the robot arm for abrasion or wear — requires opening the upper arm cover panel
- Verify TCP (tool centre point) accuracy using the calibration sphere and reference test — deviation must be below 0.1 mm
- Check backup battery voltage — replace if below 3.2V
- Review the fault log for recurring minor faults that may indicate developing issues

Every 6 Months (or per cycle count threshold)

- Full gear reducer grease replacement for J1 to J4 — performed by equipment engineering
- Full gear reducer grease replacement for J5 and J6 — higher frequency due to smaller reducer size — every 3 months or 3,000,000 cycles
- Encoder cable inspection and connector re-termination if any intermittent faults were recorded
- Full safety system verification by a qualified safety engineer — all zone boundaries, E-stop response times, and brake engagement times must be formally recorded

---

6. Safety Precautions

- **The work cell must be physically secured** before any manual operation or maintenance. Use lockable safety gates — do not rely on area scanners alone during maintenance.
- **All personnel entering the work cell** must first place their personal lock on the energy isolation point. No exceptions — including supervisors and engineers.
- **Never** defeat or bypass any safety interlock, area scanner, or light curtain — even temporarily. This is a terminable safety violation.
- **Never** exceed the rated payload of 20 kg at the tool flange. Exceeding payload affects repeatability, causes overtemperature faults, and can cause the robot to drop loads unexpectedly.
- **Never** teach robot positions with personnel inside the work cell other than the operator holding the teach pendant with the deadman switch engaged.
- The teach pendant deadman switch must be held in the intermediate position during all manual robot motion — releasing it or squeezing it fully both halt the robot. This is intentional.
- In the event of a person being struck by or trapped by the robot arm — press the nearest emergency stop immediately. Do not attempt to move the robot to free the person unless trained to do so. Call emergency services.
- All maintenance activities on the robot arm must be performed with the main isolator locked out and the stored energy in the servo drives fully discharged — wait a minimum of 5 minutes after power-off before opening the controller cabinet.

---

7. Spare Parts Reference

Part Name	Part Number	Recommended Stock
RC-7 controller backup battery	RA6200-BB-007	2 units
Controller cabinet cooling fan	RA6200-CF-006	2 units
Controller cabinet air filter foam	RA6200-AF-008	4 units

J1/J2/J3 gear reducer grease (500g)	RA6200-GR-001	2 units
J4/J5/J6 gear reducer grease (200g)	RA6200-GR-002	2 units
Teach pendant cable assembly	RA6200-TP-003	1 unit
Encoder cable set (full arm)	RA6200-EC-004	1 set
Tool flange bearing	RA6200-TF-005	1 unit
